

Total marks – 120

Attempt Questions 1 – 8

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

**Question 1 (15 marks)** Use a separate piece of paper

Marks

a) Find  $\int_0^{\pi} \frac{\sin \theta d\theta}{\cos^2 \theta}$

2

b) By completing the square, evaluate  $\int_{-1}^0 \frac{dx}{\sqrt{1-2x-x^2}}$

3

c) (i) Find  $A$ ,  $B$  and  $C$  if  $\frac{16}{(x^2+4)(2-x)} = \frac{Ax+B}{(x^2+4)} + \frac{C}{(2-x)}$

2

(ii) Hence find  $\int \frac{16}{(x^2+4)(2-x)} dx$

2

d) Use the substitution  $x = 3\sin \theta$  to find  $\int \frac{x^2}{\sqrt{9-x^2}} dx$

3

e) Using  $t = \tan \frac{x}{2}$  find  $\int \frac{dx}{1+\sin x}$

3

**Question 2 (15 marks)** Use a separate piece of paper

a) Let  $z = 2 - i$  and  $\omega = 1 + 3i$ , find;

(i)  $z\bar{\omega}$

1

(ii)  $\frac{2}{z}$

1

b) Let  $\alpha = \frac{4i}{-1+i\sqrt{3}}$

(i) Express  $\alpha$  in modulus argument form

2

(ii) Express  $\alpha^5$  in modulus argument form

2

(iii) Hence express  $\alpha^5$  in the form  $x + iy$

1

c) Sketch the region on the Argand diagram where the inequalities

3

$$\frac{\pi}{4} \leq \arg(z-i) \leq \frac{3\pi}{4} \quad \text{and} \quad |z-i| \leq 2$$

hold simultaneously

**Question 2 (continued)**

Marks

d) The point  $A$  represents the real number  $z_1 = 1$ . The point  $B$  represents the complex number  $z_2 = (1 + \sqrt{3}) + i$ . If  $ABCD$  is a square in anti-clockwise rotational order, find;

(i) the complex number  $z_3$  represented by the point  $D$ .

2

(ii) the complex number  $z_4$  represented by the point  $C$ .

2

(iii) the complex number  $z_5$  represented by the point of intersection of the diagonals  $AC$  and  $BD$ .

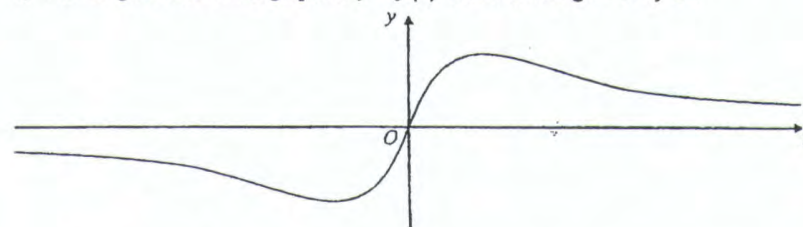
1

**Question 3 (15 marks)** Use a separate piece of paper

a) Sketch  $y = \frac{3x^2}{4-x^2}$  showing all asymptotes.

3

b) The diagram shows the graph of  $y = f(x)$  which has range  $-1 \leq y \leq 1$



Draw separate one-third page sketches of the graphs of the following;

(i)  $y = |f(x)|$

1

(ii)  $y = f(x+1)$

1

(iii)  $y = [f(x)]^2$

2

(iv)  $y = \frac{1}{f(x)}$

2

(v)  $y = e^{f(x)}$

2

c) It is known that  $P(x) = x^4 - 4x^3 + 5x^2 + ax + b$  is divisible by  $(x-2)^2$

(i) Find the values of  $a$  and  $b$ .

3

(ii) Factorise  $P(x)$  into linear factors

1

**Question 4 (15 marks)** Use a separate piece of paper

**Marks**

- a) A curve has the equation  $x^3 + 2xy - 4y^2 = 10$ .  
Find an expression for  $\frac{dy}{dx}$  as a function of  $x$  and  $y$ . 2
- b) A cylindrical hole of radius 1 cm is bored through the centre of a sphere of radius 3 cm. Using the method of cylindrical shells, calculate the exact volume of the sphere that remains. 4
- c) For the conic  $\frac{x^2}{25} + \frac{y^2}{16-\lambda} = 1$  find ;  
(i) the values of  $\lambda$  that makes the conic an ellipse with the foci on the  $x$  axis. 2  
(ii) the values of  $\lambda$  that makes the conic a rectangular hyperbola. 1  
(iii) the coordinates of the foci and the equations of the directrices when  $\lambda = 0$  2
- d) How many different ways are there of seating four married couples at a circular table with men and women in alternate positions and no wife next to her husband? 4

**Question 5 (15 marks)** Use a separate piece of paper

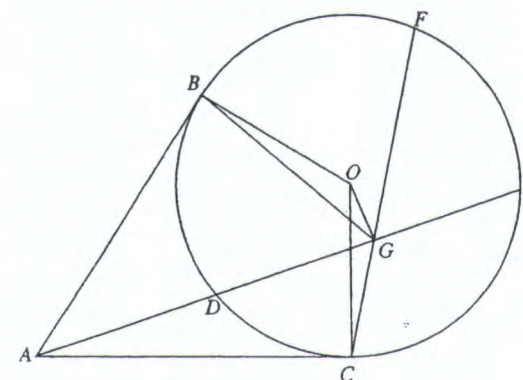
- a) Suppose  $\alpha, \beta$  and  $\gamma$  are the three roots of the polynomial equation  $x^3 + x + 12 = 0$   
(i) Find  $\alpha^2 + \beta^2 + \gamma^2$  2  
(ii) Hence explain why only one of the roots is real 1  
(iii) The real root is denoted by  $\alpha$ . Prove that  $-3 < \alpha < -2$  1  
(iv) Hence prove that the modulus of each of the other roots lies between 2 and  $\sqrt{6}$  3
- b) A woman of mass  $M$  kg jumps vertically (feet first) from a rock ledge into a river below.  
When she is falling at  $v$  m/s, she encounters air resistance equal to  $\frac{Mv}{10}$  Newtons.  
She hits the water at a speed of  $V$  m/s.  
Let  $x$  be the displacement below the rock ledge at time  $t$  seconds after jumping.  
(i) Show that  $\ddot{x} = g - \frac{v}{10}$ , where  $g$  is the acceleration due to gravity 1  
(ii) If it takes one second for her feet to hit the water, using  $g = 10 \text{ m/s}^2$  show that;  
$$V = 100 \left( 1 - e^{-\frac{1}{10}} \right)$$
  
(iii) Find the height of the rock ledge above the water, to the nearest 0.1 metre 4

**Question 6 (15 marks)** Use a separate piece of paper

**Marks**

- a) By considering the series  $1 + t + t^2 + t^3 + \dots + t^n$ , or otherwise;  
(i) sum the series  $1 + 2t + 3t^2 + 4t^3 + \dots + nt^{n-1}$  2  
(ii) hence evaluate  $1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \dots + 20 \times 2^{19}$  1

b)



In the diagram,  $AB$  and  $CA$  are the tangents from  $A$  to the circle with centre  $O$ , meeting the circle at  $B$  and  $C$ .

$ADE$  is a secant of the circle.  $G$  is the midpoint of  $DE$ .  $CG$  produced meets the circle at  $F$ .

(Note: on page 10 there is a copy of the above diagram that you can use, please detach and include with your solutions)

- (i) Show that  $ABOC$  is a cyclic quadrilateral 2  
(ii) Show that  $AOGC$  is a cyclic quadrilateral 2  
(iii) Hence prove that  $BF \parallel AE$  3

Question 6 (continued)

Marks

- c) The roof of a sports stadium has an elliptical base with a major axis of length  $2a$  and minor axis of length  $2b$ . The two identical sloping tops are inclined at  $30^\circ$  to the base.

$PQRS$  represents a rectangular cross-section of thickness  $\delta x$  taken  $x$  units from the centre  $O$  of the ellipse.

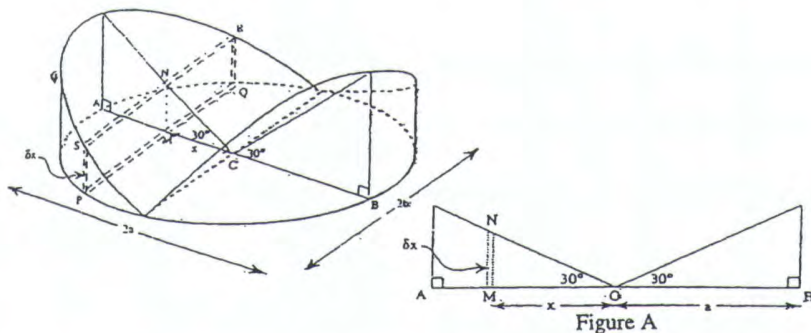


Figure A shows a side view of the stadium if sliced in half along  $AB$

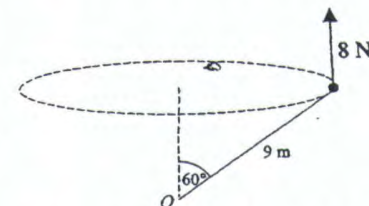
- Find the height  $MN$  of the rectangular cross-section  $PQRS$  1
- Show that the area  $A$  of the rectangle  $PQRS$  is given by  $A = \frac{2b}{a\sqrt{3}} x\sqrt{a^2 - x^2}$  2
- Calculate the volume of the stadium roof. 2

24

Question 7 (15 marks) Use a separate piece of paper

Marks

a)



A toy aircraft of mass  $0.5 \text{ kg}$  is attached to one end of a string of length  $9 \text{ m}$ . The other end of the string is attached to a fixed point  $O$  and moves with constant angular velocity.

The string is taut, and makes an angle of  $60^\circ$  with the upward vertical at  $O$ .

In a simplified model of the motion, the aircraft is treated as a particle and the force of the air on the aircraft is taken to act vertically upwards with magnitude  $8 \text{ Newtons}$ . (Use  $g = 10 \text{ m/s}^2$ )

- Resolve the forces on the aircraft in the horizontal and vertical directions 2
- Find the tension in the string 1
- Find the speed of the aircraft in  $\text{m/s}$ . 2

b)  $P\left(cp, \frac{c}{p}\right)$  is a point on the hyperbola  $xy = c^2$

- Show that the equation of the tangent at  $P$  has the equation  $x + p^2y = 2cp$  2
- The tangent at  $P$  cuts the  $x$  axis and  $y$  axis at  $A$  and  $B$  respectively. Find the coordinates of  $A$  and  $B$ . 1
- $Q$  is the fourth vertex of the rectangle  $OAQB$ . Show that the locus of  $Q$  is another rectangular hyperbola. 2

c) (i) Let  $I_n = \int \tan^n \theta d\theta$ , show that  $I_n = \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}$  3

(ii) Hence find the exact value of  $\int_0^{\frac{\pi}{4}} \tan^4 \theta d\theta$  2



G(4)  $P(x) = x^3 - 7x^2 + 10x + a$   
 $P(x) = 4x^3 - 12x^2 + 10x + a$

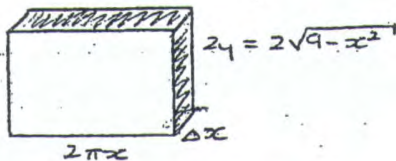
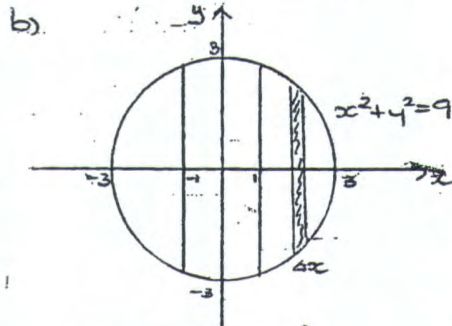
$P(2) = 0$   
 $16 - 32 + 20 + 2a + b = 0$   
 $2a + b = -4$

$P'(2) = 0$   
 $32 - 48 + 20 + a = 0$   
 $a = -4, \therefore b = 4$  (3)

(iii)  $P(x) = x^4 - 4x^3 + 5x^2 - 4x + 4$   
 $= (x^2 + 4x + 4)(x^2 + 1)$   
 $= (x+2)^2(x+i)(x-i)$  (1)

#### Question 4 (15)

a)  $x^3 + 2xy - 4y^2 = 10$   
 $3x^2 + (2x)\frac{dy}{dx} + (y)(2) - 8y\frac{dy}{dx} = 0$   
 $(2x - 6y)\frac{dy}{dx} = -3x^2 - 2y$   
 $\frac{dy}{dx} = \frac{3x^2 + 2y}{6y - 2x}$  (2)



$A(x) = 4\pi x \sqrt{9-x^2}$   
 $\Delta V = 4\pi x \sqrt{9-x^2} \Delta x$   
 $V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^3 4\pi x \sqrt{9-x^2} \Delta x$   
 $= 2\pi \int_1^3 2x \sqrt{9-x^2} dx$   
 $= -2\pi \left[ \frac{2}{3}(9-x^2)^{3/2} \right]_1^3$   
 $= -\frac{4\pi}{3}(0 - 8\sqrt{8})$  (4)

(i)  $-16 < \lambda < 9$   
 $-9 < \lambda < 16$  (2)  
(ii)  $16 - \lambda = -25$   
 $\lambda = 41$  (1)  
(iii)  $\frac{x^2}{25} + \frac{y^2}{16} = 1$   
 $a^2 = 25, b^2 = 16$   
 $a = 5, b = 4$   
 $1 - e^2 = \frac{b^2}{a^2} = \frac{16}{25}$   
 $e^2 = \frac{9}{25}$   
 $e = \frac{3}{5}$  (2)  
 $\therefore$  foci  $(\pm 3, 0)$ , directrices  $x = \pm \frac{25}{3}$

d) Place men first = 3!  
 A Wife A has 2 spots  
 B D this leaves 1 spot for  
 C (M) each of the other  
 wives  
 Ways =  $3! \times 2 \times 1 \times 1 \times 1$   
 $= 12$  (4)

#### Question 5 (15)

a) (i)  $x^3 + x + 12 = 0$   
 $\sum x^2 = (\sum x)^2 - 2\sum x\beta$   
 $= 0^2 - 2(1)$   
 $= -2$  (2)  
(ii) as  $\sum x^2 < 0$  there must be an imaginary root.  
 However when coefficients are real, roots appear in conjugate pairs  
 $\therefore$  there are two imaginary roots  
 $\therefore$  there is one real root (1)

(iii)  $P(-3) = -27 - 3 + 12 = -18 < 0$   
 $P(-2) = -8 - 2 + 12 = 2 > 0$   
 as polynomial is continuous  
 $-3 < \alpha < -2$  (1)

(iv)  $\alpha\beta\gamma = -12$   
 $4 < \beta\gamma < 6$   
 $\beta\gamma$  are conjugates  
 $z\bar{z} = |z|^2$   
 $\therefore 4 < |z|^2 < 6$   
 $2 < |z| < \sqrt{6}$  (3)

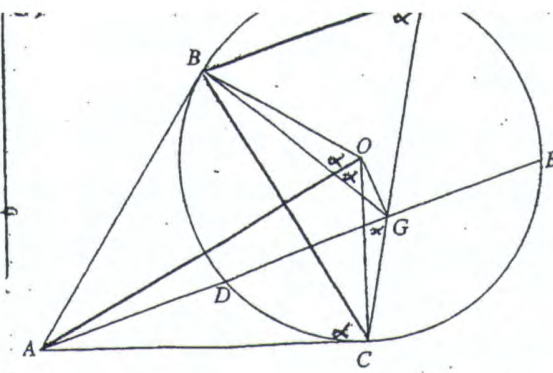
25  
 $M\ddot{x} = Mg - \frac{mv}{10}$   
 $\ddot{x} = g - \frac{v}{10}$  (1)  
(ii)  $\frac{dv}{dt} = \frac{10g - v}{10}$   
 $t = -10 \int \frac{-dv}{10g - v}$   
 when  $t = 1, v = V$   
 $1 = -10 \int_0^V \frac{-dv}{10g - v}$   
 $-\frac{1}{10} = [\log(10g - v)]_0^V$   
 $= \log\left(\frac{10g - v}{10g}\right)$   
 $e^{-\frac{1}{10}} = \frac{10g - v}{10g}$   
 $\log e^{-\frac{1}{10}} = \log\left(\frac{10g - v}{10g}\right)$   
 $V = 10g(1 - e^{-\frac{1}{10}})$  (3)  
 $V = 100(1 - e^{-\frac{1}{10}})$  ( $g = 10$ )

(iii)  $v \frac{dv}{dx} = \frac{10g - v}{10}$   
 $x = 10 \int \frac{v dv}{10g - v}$   
 $x = -10 \int \left(1 - \frac{10g}{10g - v}\right) dv$   
 $= -10 \left[ v + 10 \log(10g - v) \right]_0^V$   
 $= -10 \left( V + 10 \log\left(\frac{10g - V}{10g}\right) \right)$   
 $= -10 \left( 100(1 - e^{-\frac{1}{10}}) + 100(-\frac{1}{10}) \right)$   
 $= 4.8 \text{ m}$  (4)

#### Question 6 (15)

a) (i)  $1 + t + t^2 + \dots + t^n = \frac{t^{n+1} - 1}{t - 1}$   
 $t + 2t + 3t^2 + \dots + nt^n = \frac{(t-1)(n+1)t^{n+1} - (1^{n+1} - 1)}{(t-1)^2}$   
 $= \frac{(n+1)t^{n+1} - n t^{n+1} + 1}{(t-1)^2}$  (2)

(ii)  $t = 2, n = 20$   
 $1 + 2 \times 2 + 3 \times 2^2 + \dots + 20 \times 2^{19}$   
 $= 19 \times 2^{20} - 20 \times 2^{19} + 1$   
 $= 9427145$  (1)



(i)  $\angle OBA = 90^\circ$  (radius  $\perp$  tangent)  
 $\angle OCA = 90^\circ$  ( " )  
 $\therefore$  ABOC is cyclic quadrilateral (2)  
 (opposite  $\angle$ 's supplementary)  
(ii)  $\angle OCA = 90^\circ$  ( $\perp$  centre bisects chord)  
 $\therefore \angle OCA = \angle OBA$   
 AOCB is cyclic quadrilateral (2)  
 ( $\angle$ 's in same segment =)

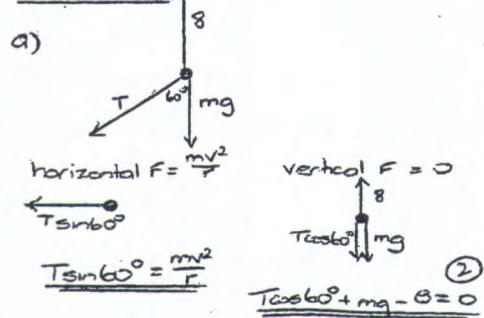
(iii) Let  $\angle BFC = x$   
 $\angle BOC = 2x$  (Lat centre twice  
 Lat circumference on  
 same arc)  
 $\angle BCA = x$  (alt in alternate segment)  
 $\therefore \angle BOA = x$  ( $\angle$ 's in same segment =  
 in ABOC)  
 $\angle BOC = \angle AOB + \angle AOC$  (common  $\angle$ )  
 $2x = x + \angle AOC$   
 $\angle AOC = x$   
 $\angle AOC = \angle ACC$  ( $\angle$ 's in same segment =  
 in AOC)  
 $\therefore \angle AGC = x$   
 $\angle AGC = \angle BFC = x$  (3)  
 $\therefore BF \parallel AE$  (corresponding  $\angle$ 's =)

c)  $\frac{MN}{x} = \tan 30^\circ$   
 $\therefore MN = \frac{x}{\sqrt{3}}$  (1)

(ii) 
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$   
 $a^2 y^2 = a^2 b^2 - b^2 x^2$   
 $y^2 = \frac{b^2(a^2 - x^2)}{a^2}$   
 $A(x) = \frac{x}{\sqrt{3}} \times \frac{2b\sqrt{a^2 - x^2}}{2}$   
 $= \frac{2b}{\sqrt{3}} x \sqrt{a^2 - x^2}$  (2)

$$\begin{aligned}
 \text{iii) } \Delta V &= \frac{2b}{\sqrt{3}} \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx \\
 V &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n \frac{2b}{\sqrt{3}} \frac{1}{\sqrt{a^2-x_i^2}} \Delta x \\
 &= \frac{2b}{\sqrt{3}} \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx \\
 &= \frac{2b}{\sqrt{3}} \left[ \frac{1}{a} \sin^{-1} \frac{x}{a} \right]_0^a \\
 &= \frac{2b}{\sqrt{3}} \left( \frac{1}{a} \sin^{-1} 1 \right) \\
 &= \frac{2b}{\sqrt{3}} \left( \frac{1}{a} \cdot \frac{\pi}{2} \right) \\
 &= \frac{\pi b}{\sqrt{3} a} \quad (2)
 \end{aligned}$$

Question 7 (15)



ii)  $\frac{T}{2} = 8 - (0.5)(10)$

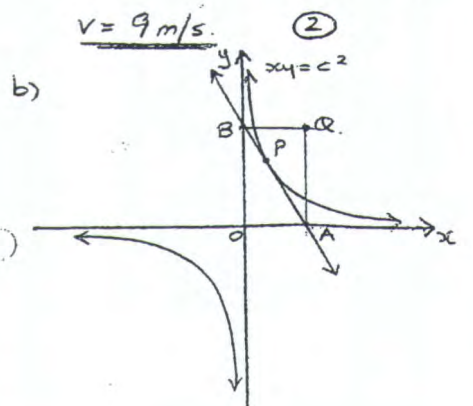
$T = 6 \text{ N}$  (1)

iii)  $\frac{\sqrt{3}T}{2} = \frac{mv^2}{r}$

$\frac{\sqrt{3}(6)}{2} = \frac{(0.5)v^2}{\frac{9\sqrt{3}}{2}}$

$v^2 = 3\sqrt{3} \times \frac{9\sqrt{3}}{2} \times 2$

$v = 9 \text{ m/s}$  (2)



iv)  $y = x$

$\frac{dy}{dx} = \frac{-c^2}{x^2}$

when  $x = cp$ ,  $\frac{dy}{dx} = -\frac{1}{p^2}$

$y - \frac{c}{p} = -\frac{1}{p^2} (x - cp)$

$p^2y - pc = -x + cp$

$x + p^2y = 2cp$  (2)

A:  $y = 0$  B:  $x = 0$

$x = 2cp$   $p^2y = 2cp$

$y = \frac{2c}{p}$

B:  $(0, \frac{2c}{p})$  (1)

iii) Q  $(2cp, \frac{2c}{p})$

$xy = 2cp \times \frac{2c}{p}$

$xy = 4c^2$

$\therefore$  locus of Q is  $xy = 4c^2$  (2)

c) i)  $I_n = \int \tan^n \theta d\theta$

$= \int \tan^{n-2} \theta (\sec^2 \theta - 1) d\theta$

$= \int \tan^{n-2} \theta \sec^2 \theta d\theta - \int \tan^{n-2} \theta d\theta$

$= \int u^{n-2} du - I_{n-2}$   $u = \tan \theta$   $du = \sec^2 \theta d\theta$

$= \frac{u^{n-1}}{n-1} - I_{n-2}$

$= \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}$  (3)

ii)  $\int_0^{\frac{\pi}{4}} \tan^4 \theta = I_4$

$= \left[ \frac{1}{3} \tan^3 \theta \right]_0^{\frac{\pi}{4}} - I_2$

$= \left[ \frac{1}{3} \tan^3 \theta - \tan \theta \right]_0^{\frac{\pi}{4}} + I_0$

$= \left[ \frac{1}{3} \tan^3 \theta - \tan \theta + \theta \right]_0^{\frac{\pi}{4}}$

$= \frac{1}{3} - 1 + \frac{\pi}{4}$

$= \frac{\pi}{4} - \frac{2}{3}$

$= \frac{3\pi - 8}{12}$  (2)

a) i)  $y = x \sin x$

$\frac{dy}{dx} = (x)(\cos x) + (\sin x)(1)$

$= x \cos x + \sin x$

$\frac{d^2y}{dx^2} = (x)(-\sin x) + (\cos x)(1) + \cos x$

$= 2 \cos x - x \sin x$  (1)

ii) Prove true for  $n=1$

LHS =  $\frac{d^2y}{dx^2} = 2 \cos x - x \sin x$

RHS =  $(-1)(x \sin x - 2 \cos x) = 2 \cos x - x \sin x$

$\therefore$  LHS = RHS

Hence the result is true for  $n=1$

Assume the result is true for  $n=k$

$\therefore \frac{d^{2k}y}{dx^{2k}} = (-1)^k (x \sin x - 2k \cos x)$

Prove the result is true for  $n=k+1$

$\therefore \frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^{k+1} (x \sin x - 2(k+1) \cos x)$

Proof:

$\frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^k \left[ (x)(\cos x) + \sin x + 2 \cos x \right]$

$= (-1)^k (x \cos x + (2k+2) \sin x)$

$\frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^k \left[ (x)(-\sin x) + (\cos x)(1) + (2k+2) \cos x \right]$

$= (-1)^k (-x \sin x + (2k+2) \cos x)$

$= (-1)^{k+1} (x \sin x - 2(k+1) \cos x)$

Hence the result is true for  $n=k+1$  if it is also true for  $n=k$

Hence the result is true for all positive integral values of  $n$  by induction. (3)

b) i)  $e^{-x} \sin x = 0$

$\sin x = 0$

$x = 0, \pi, 2\pi, \dots$

$\therefore P(0,0), Q(2\pi,0), R(4\pi,0)$  (1)

ii)  $\int e^{-x} \sin x dx$

$u = e^{-x}$   $v = -\cos x$

$du = -e^{-x} dx$   $dv = \sin x dx$

$= -e^{-x} \cos x - \int e^{-x} \cos x dx$

$u = e^{-x}$   $v = \sin x$

$du = -e^{-x} dx$   $dv = \cos x dx$

$\therefore 2 \int e^{-x} \sin x dx = -e^{-x} (\sin x + \cos x) + C$

$\int e^{-x} \sin x dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + C$  (3)

iii)  $A_1 = \int_0^{\pi} e^{-x} \sin x dx$

$A_2 = \int_{\pi}^{2\pi} e^{-x} \sin x dx$

$\therefore A_n = \int_{(n-1)\pi}^{n\pi} e^{-x} \sin x dx$

$= -\frac{1}{2} \left[ e^{-x} (\sin x + \cos x) \right]_{(n-1)\pi}^{n\pi}$

$= -\frac{1}{2} \left( e^{-n\pi} (\sin n\pi + \cos n\pi) - e^{-(n-1)\pi} (\sin (n-1)\pi + \cos (n-1)\pi) \right)$

$= \frac{1}{2} (e^{-(2n-2)\pi} + e^{-(1-2n)\pi})$  (2)

iv)  $A_1 = \frac{1}{2} (e^{-\pi} + 1)$

$A_2 = \frac{1}{2} (e^{-3\pi} + e^{-2\pi})$

$A_3 = \frac{1}{2} (e^{-5\pi} + e^{-4\pi})$

$\therefore$  geometric series  $r = e^{-2\pi}$

$S_{\infty} = \frac{\frac{1}{2} (e^{-\pi} + 1)}{1 - e^{-2\pi}}$

$= \frac{e^{-\pi} + 1}{2(e^{\pi} - 1)}$

$= \frac{e^{\pi} + 1}{2(e^{\pi} - 1)}$  (3)

$A_n$  = areas above  $x$  axis

v) Let  $A_1 + A_2 + A_3 + \dots = X$

$a_n$  = areas under  $x$  axis

$a_1 + a_2 + a_3 + \dots = Y$

$\therefore \int_0^{\infty} e^{-x} \sin x dx = X - Y$

$\frac{1}{2} = \frac{e^{\pi}}{2(e^{\pi} - 1)} - Y$

$Y = \frac{e^{\pi} - (e^{\pi} - 1)}{2(e^{\pi} - 1)}$

$= \frac{1}{2(e^{\pi} - 1)}$

$\int_0^{\infty} |e^{-x} \sin x| dx = X + Y$

$= \frac{e^{\pi}}{2(e^{\pi} - 1)} + \frac{1}{2(e^{\pi} - 1)}$

$= \frac{e^{\pi} + 1}{2(e^{\pi} - 1)}$  (2)